

Project Design Phase
Proposed Solution Template

Date	26 June 2026
Team ID	Solo
Project Name	ShopEZ Ecommerce Application
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Small-to-medium retail merchants lack an affordable, easy-to-use platform that combines a professional online storefront with real-time business management tools, forcing them to rely on manual processes (notebooks, WhatsApp) or expensive/generic marketplaces that offer no brand identity, while customers are left with impersonal, cluttered buying experiences.
2.	Idea / Solution description	ShopEZ is a full-stack MERN e-commerce application offering customers a premium "boutique" shopping experience — live search & category filters, multi-tab product details, a live-syncing cart, and a 3-step checkout (Card/UPI/COD) — paired with a Seller Dashboard for inventory CRUD, order-status management, and live KPI analytics (revenue, total/pending/delivered orders).
3.	Novelty / Uniqueness	Combines a premium "Virtual Closet/Art Gallery" boutique aesthetic (not a generic grid template) with a unified seller-facing analytics dashboard in one application — most small-seller tools offer either a plain storefront or basic order lists, not both in an elegant, integrated package.
4.	Social Impact / Customer Satisfaction	Empowers small and medium merchants to compete online without high commissions or technical barriers, helping local businesses grow; customers get a trustworthy, transparent shopping journey with real-time order tracking and multiple payment choices, increasing satisfaction and trust.
5.	Business Model (Revenue Model)	Can be monetized via a subscription/tiered plan for sellers (basic listing vs. premium analytics/inventory features), a small transaction/commission fee per completed order, or optional paid add-ons (featured listings, promoted products, premium storefront themes).

6.	Scalability of the Solution	Built on the MERN stack with MongoDB Atlas (cloud-hosted) and a stateless JWT auth system, deployed on Render via a unified render.yaml blueprint — this architecture supports horizontal scaling of the Express API, easy onboarding of new sellers/categories, and straightforward extension to mobile apps or additional payment gateways as demand grows.
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1 CUSTOMER SEGMENT(S) Small-to-medium retail merchants needing an easy online storefront, and value-conscious online shoppers who want a trustworthy, curated shopping experience with flexible payment options.	2 JOBS-TO-BE-DONE / PROBLEMS Merchants: launch a professional online store and manage inventory & orders without heavy cost/tech. Customers: browse, compare and buy products smoothly, track orders, and pay via CardUPI/COD.	3 TRIGGERS A seller sees competitors selling online while their shop stays offline and loses customers. A shopper abandons a cluttered, slow app during checkout and looks for a simpler alternative.	4 EMOTIONS: BEFORE / AFTER Before: frustration and anxiety from lost sales and manual, error-prone order tracking. After: confidence and relief with a real-time dashboard, live KPIs and a reliable checkout flow.
5 AVAILABLE SOLUTIONS (AS) Shopify/WooCommerce: powerful but costly and technical for small sellers to set up and run. Amazon/Flipkart marketplaces: high commissions, no brand identity or direct customer data. Manual selling via WhatsApp/ phone calls: unscaleable, error-prone, no analytics or tracking.	9 PROBLEM ROOT CAUSE No affordable, easy-to-use dedicated platform combines an elegant, boutique-style storefront with real-time seller analytics and inventory tools — so small merchants fall back on fragmented, manual processes (notebooks, WhatsApp) to run their shop, while shoppers get generic, impersonal buying experiences.		6 CUSTOMER CONSTRAINTS (CC) Limited budget for subscription fees or high commissions. Limited technical skill to build or maintain a website. Need mobile-friendly access and secure, multiple payment modes (Card, UPI, Cash on Delivery).
7 BEHAVIOUR (BE) Sellers track orders manually in notebooks/spreadsheets or via WhatsApp chats with customers. Shoppers browse multiple apps, compare price/reviews across platforms before purchasing.	10 YOUR SOLUTION ShopEZ — a full-stack MERN e-commerce application. Customers get a premium "boutique" shopping experience: live search & category filters, multi-tab product details, a live-syncing cart, and a 3-step checkout supporting Card, UPI and Cash on Delivery. Sellers get a dedicated dashboard: live KPI cards (Revenue, Total/Pending/Delivered Orders), full inventory CRUD, and order-status management (Pending → Processing → Shipped → Delivered → Cancelled). Secured with stateless JWT auth, deployed on Render (Node 20.15.0). <i>Problem-Solution Fit: ShopEZ solves merchants' need for an affordable, easy storefront + analytics, and shoppers' need for a trustworthy, premium buying experience — closing the gap left by boxes 5 & 6.</i>		8 CHANNELS OF BEHAVIOUR WhatsApp Business & Instagram/ Facebook shop pages. Third-party marketplaces (Amazon, Flipkart, Meesho). Offline retail counters and word-of-mouth referrals.